EECS 116/216 Project 1 - Team 2

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1 Introduction

The purpose of this project is to estimate the state of a micro aerial vehicle by implementing an Extended Kalman Filter (EKF). By fusing Inertial Measurement Unit (IMU) data with partial motion capture (mocap) observations, the filter estimates the vehicle's pose, velocity, and sensor biases. The process and measurement noise covariances were tuned to obtain accurate estimates.

2 Methodology

We estimate the state $\mathbf{x} = [\mathbf{x}_1\ \mathbf{x}_2\ \mathbf{x}_3\ \mathbf{x}_4\ \mathbf{x}_5]^T = [\mathbf{p}\ \mathbf{q}\ \dot{\mathbf{p}}\ \mathbf{b}_g\ \mathbf{b}_a]^T \in \mathbb{R}^{15}$, whose components are the world-frame position, body-to-world orientation (roll, pitch, yaw), world-frame velocity, and body-frame gyroscope and accelerometer biases. The inputs are the gyroscope and accelerometer measurements, $\mathbf{u} = [\boldsymbol{\omega}_m\ \mathbf{a}_m]^T \in \mathbb{R}^6$.

We assume the body, IMU, and tracked mocap frames and their origins coincide. Since the accelerometer measures specific force rather than kinematic acceleration, the measurements are modeled as

$$\begin{aligned} \boldsymbol{\omega}_m &= \boldsymbol{\omega} + \mathbf{b}_g + \mathbf{n}_g, & \mathbf{n}_g &\sim \mathcal{N}(\mathbf{0}, Q_g), \\ \mathbf{a}_m &= R(\mathbf{q})^T(\ddot{\mathbf{p}} - \mathbf{g}) + \mathbf{b}_a + \mathbf{n}_a, & \mathbf{n}_a &\sim \mathcal{N}(\mathbf{0}, Q_a), \end{aligned} \tag{1}$$

where $R = R_z(\text{yaw})R_y(\text{pitch})R_x(\text{roll})$ is the intrinsic ZYX rotation matrix and $\mathbf{g} = [0, 0, -9.8]^T\ \text{m/s}^2$. The bias drift is modeled as white Gaussian noise, $\dot{\mathbf{b}}_g = \mathbf{n}_{bg} \sim \mathcal{N}(\mathbf{0}, Q_{bg})$ and $\dot{\mathbf{b}}_a = \mathbf{n}_{ba} \sim \mathcal{N}(\mathbf{0}, Q_{ba})$.

The gyroscope measures body angular velocity, not Euler-angle rates. The two are related by $\boldsymbol{\omega} = G(\mathbf{q})\dot{\mathbf{q}}$, where

$$G = \begin{bmatrix} 1 & 0 & -\sin(\text{pitch}) \\ 0 & \cos(\text{roll}) & \sin(\text{roll})\cos(\text{pitch}) \\ 0 & -\sin(\text{roll}) & \cos(\text{roll})\cos(\text{pitch}) \end{bmatrix}. \tag{2}$$

The resulting process model is

$$\dot{\mathbf{x}} = \begin{bmatrix} \mathbf{x}_3 \\ G(\mathbf{x}_2)^{-1}(\boldsymbol{\omega}_m - \mathbf{x}_4 - \mathbf{n}_g) \\ \mathbf{g} + R(\mathbf{x}_2)(\mathbf{a}_m - \mathbf{x}_5 - \mathbf{n}_a) \\ \mathbf{n}_{bg} \\ \mathbf{n}_{ba} \end{bmatrix} = f(\mathbf{x}, \mathbf{u}, \mathbf{n}), \tag{3}$$

where $\mathbf{n} = [\mathbf{n}_g\ \mathbf{n}_a\ \mathbf{n}_{bg}\ \mathbf{n}_{ba}]^T \sim \mathcal{N}(\mathbf{0}, Q)$ and Q is the process noise covariance. The EKF prediction step with forward Euler integration is

$$\begin{aligned} \bar{\mu}_t &= \mu_{t-1} + \delta t\, f(\mu_{t-1}, u_{t-1}, 0), \\ \bar{\Sigma}_t &= F_t \Sigma_{t-1} F_t^T + U_t Q_d U_t^T, \end{aligned} \tag{4}$$

with $F_t = I + \delta t\, A_t$, $A_t = \partial f/\partial \mathbf{x}|_{\mu_{t-1},u_{t-1},0}$, $U_t = \partial f/\partial \mathbf{n}|_{\mu_{t-1},u_{t-1},0}$, and $Q_d = Q\,\delta t$. The nonzero blocks of the Jacobians are

$$\begin{aligned}
A[0{:}3, 6{:}9] &= I_3 & U[3{:}6, 0{:}3] &= -G(\mathbf{x}_2)^{-1} \\
A[3{:}6, 3] &= \frac{\partial (G^{-1})}{\partial \text{roll}}(\boldsymbol{\omega}_m - \mathbf{x}_4) & U[6{:}9, 3{:}6] &= -R \\
A[3{:}6, 4] &= \frac{\partial (G^{-1})}{\partial \text{pitch}}(\boldsymbol{\omega}_m - \mathbf{x}_4) & U[9{:}12, 6{:}9] &= I_3 \\
A[3{:}6, 9{:}12] &= -G(\mathbf{x}_2)^{-1} & U[12{:}15, 9{:}12] &= I_3 \\
A[6{:}9, 3] &= \frac{\partial R}{\partial \text{roll}}(\mathbf{a}_m - \mathbf{x}_5) \\
A[6{:}9, 4] &= \frac{\partial R}{\partial \text{pitch}}(\mathbf{a}_m - \mathbf{x}_5) \\
A[6{:}9, 5] &= \frac{\partial R}{\partial \text{yaw}}(\mathbf{a}_m - \mathbf{x}_5) \\
A[6{:}9, 12{:}15] &= -R
\end{aligned} \tag{5}$$

Since G depends only on roll and pitch, there is no yaw column in the orientation block. The derivatives of G^{-1} are computed with the identity $\partial(G^{-1})/\partial\theta = -G^{-1}(\partial G/\partial\theta)G^{-1}$, and those of R with the product rule on $R = R_z R_y R_x$, differentiating one factor at a time.

The EKF update step is

$$\begin{aligned}
K_t &= \bar{\Sigma}_t C_t^T (C_t \bar{\Sigma}_t C_t^T + W_t \hat{R} W_t^T)^{-1}, \\
\mu_t &= \bar{\mu}_t + K_t\big(\mathbf{z}_t - g(\bar{\mu}_t, 0)\big), \\
\Sigma_t &= \bar{\Sigma}_t - K_t C_t \bar{\Sigma}_t,
\end{aligned} \tag{6}$$

where $\mathbf{z}_t = g(\mathbf{x}_t, \mathbf{v}_t)$ is the observation model, $\mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \hat{R})$ is the measurement noise, $C_t = \partial g/\partial \mathbf{x}|_{\bar{\mu}_t,0}$, and $W_t = \partial g/\partial \mathbf{v}|_{\bar{\mu}_t,0}$.

3 Implementation

3.1 Filter Loop and Initialization

The state is initialized from the first nine entries of the first column of `vicon` (pose and velocity), with zero biases and $\Sigma_0 = I_{15}$. For each later sample $t \geq 1$, we compute $\delta t = \texttt{time}[t] - \texttt{time}[t-1]$, predict using the IMU reading at sample $t-1$, update using the mocap measurement at sample t, and store the posterior mean μ_t.

3.2 Prediction Step

The prediction step, Eqs. (4)–(5), is shared by both parts. The rotation matrix is built with SciPy's intrinsic `"ZYX"` convention, with the angles passed in the order [yaw, pitch, roll]. The mean is propagated with the full nonlinear model of Eq. (3) with the noise set to zero. Because the noise is scaled as $Q_d = Q\,\delta t$, Q is interpreted as a continuous-time noise intensity. Finally, G is singular at pitch $= \pm 90°$ (gimbal lock), but pitch stays far from this value in all datasets, so no special handling was needed.

3.3 Update with Mocap Position and Orientation (Part 1)

The measurement is $\mathbf{z}_t = [\mathbf{x}_1\ \mathbf{x}_2]^T \in \mathbb{R}^6$. The observation model is linear with respect to our state and noise, so $C_t = [I_6\ \ 0_{6\times 9}]$ and $W_t = I_6$ are constant, and the EKF update reduces to the standard Kalman update. Because angles are circular, the orientation components of the innovation $\mathbf{z}_t - C_t\bar{\mu}_t$ are wrapped to $[-\pi, \pi)$ via $\Delta\theta \leftarrow ((\Delta\theta + \pi) \bmod 2\pi) - \pi$. The state angles themselves are left unwrapped.

3.4 Update with Mocap Velocity (Part 2)

The measurement is $\mathbf{z}_t = \mathbf{x}_3 \in \mathbb{R}^3$, with $C_t = [0_{3\times 6}\ \ I_3\ \ 0_{3\times 6}]$ and $W_t = I_3$ due again to the linearity with respect to our state and noise. The mocap pose is used only for initialization. Position, orientation, and the

biases are therefore never measured directly, and are corrected only through the cross-covariance terms of $\bar{\Sigma}_t$ built up by F_t in the prediction step.

3.5 Noise Tuning

Both Q and $\hat{R}$, our chosen noise covariance matrices, are diagonal and were tuned by hand through iterative trial and error, starting from identity matrices. Each candidate was judged by comparing the estimates with the mocap on all three datasets, with most attention on the channels not used in the update step (see Section 4.3). A single Q is used for both parts, since they share the same process model. The final values are listed in Table 1.

Matrix	Entries	Value
Q_g (gyroscope noise)	x, y, z	10^{-4}
Q_a (accelerometer noise)	x, y / z	10^{-2} / 2×10^{-2}
Q_{bg} (gyroscope bias drift)	x, y, z	10^{-6}
Q_{ba} (accelerometer bias drift)	x, y, z	10^{-4}
$\hat{R}$, Part 1	position	10^{-5}
	roll, pitch / yaw	1.5×10^{-3} / 2.5×10^{-5}
$\hat{R}$, Part 2	v_x, v_y / v_z	3×10^{-2} / 4×10^{-3}

Table 1: Process and measurement noise covariances used for all reported runs.

Process noise. The gyroscope noise is two orders of magnitude below the accelerometer noise, so the orientation prediction is trusted more than the velocity prediction. Both bias-drift terms are well below the sensor-noise terms, since biases change slowly. The gyroscope biases settle within a few seconds and then stay flat on all datasets (Figs. 1–3), so a very small drift (10^{-6}) suffices. The accelerometer biases are given a larger drift (10^{-4}) because they do change during a run; for example, the x-axis bias of Dataset 9 decreases steadily from about 0.2 to 0.08 $\mathrm{m/s^2}$ (Fig. 3).

Measurement noise. In Part 1, the position variance is very small because the mocap position is highly precise, and the estimate follows it almost exactly. Roll and pitch are given a larger variance than yaw, so they rely more on gyroscope integration, at the cost of small offsets during the fast maneuvers of Dataset 4 (Fig. 2). Yaw has no gravity reference, so it is held mainly by the measurement and is given a much smaller variance. In Part 2, the horizontal velocity innovation also drives the roll, pitch, and horizontal bias corrections (Section 4.3), so a larger variance on v_x, v_y keeps these corrections gradual. The vertical velocity has a smaller variance because the vertical accelerometer axis has the largest process noise, so this channel relies more on the measurement.

4 Results

4.1 Part 1: Mocap Position and Orientation

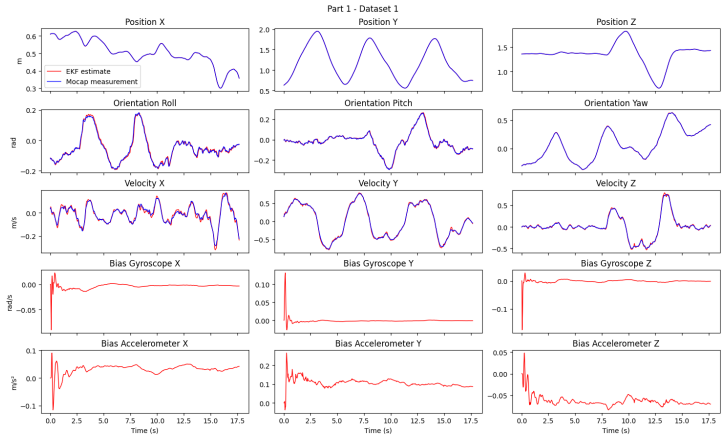


Figure 1: Part 1, Dataset 1: EKF estimates (red) and mocap measurements (blue).

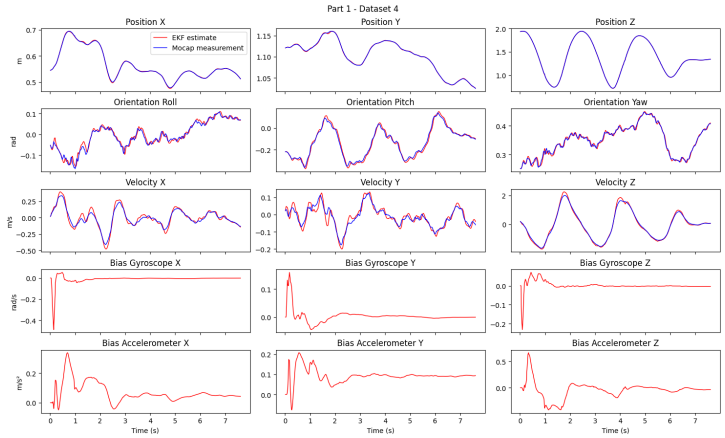


Figure 2: Part 1, Dataset 4: EKF estimates (red) and mocap measurements (blue).

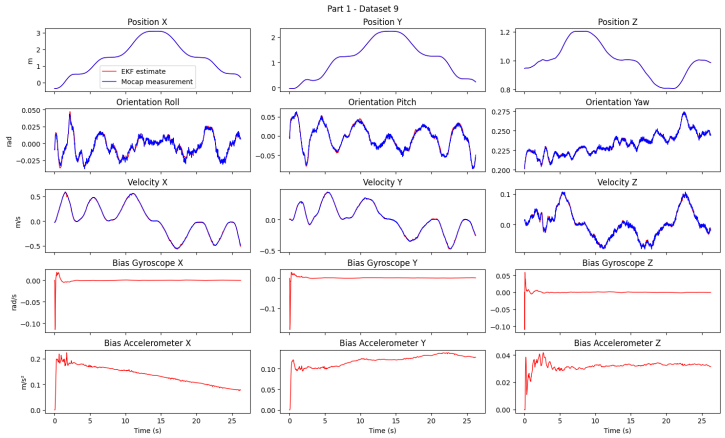


Figure 3: Part 1, Dataset 9: EKF estimates (red) and mocap measurements (blue).

4.2 Part 2: Mocap Velocity

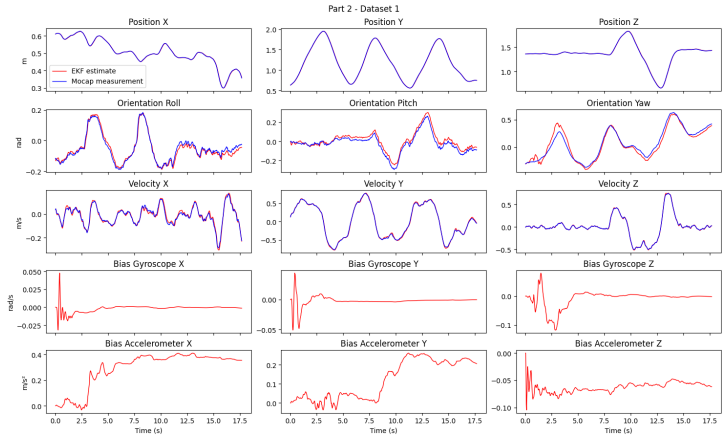


Figure 4: Part 2, Dataset 1: EKF estimates (red) and mocap measurements (blue).

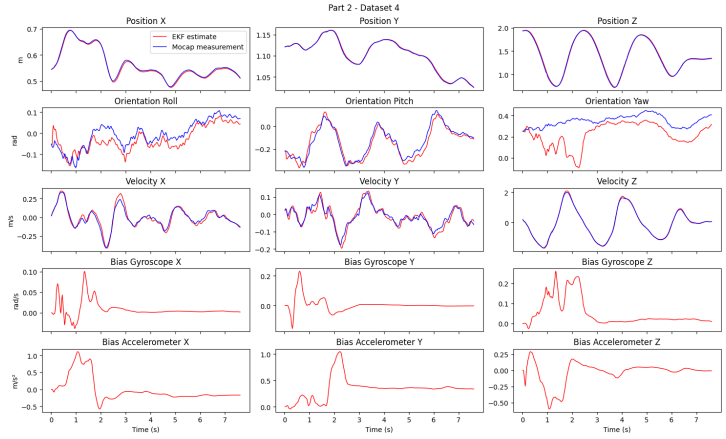


Figure 5: Part 2, Dataset 4: EKF estimates (red) and mocap measurements (blue).

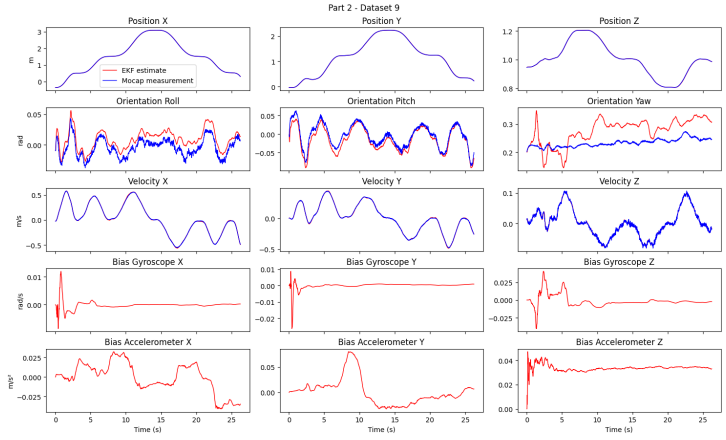


Figure 6: Part 2, Dataset 9: EKF estimates (red) and mocap measurements (blue).

4.3 Discussion

The channels used in the update step track the mocap for almost any reasonable tuning, so agreement on them is not an independent measure of accuracy. The discussion below therefore focuses on the states each part does *not* measure.

Part 1. Position and orientation overlap with the mocap on all datasets, as expected since both are measured. Velocity, which is not measured, also closely matches the mocap velocity, with only small deviations at the sharpest peaks of Dataset 4 (Fig. 2), the most aggressive run with vertical speeds up to 2 m/s. This shows that position corrections are correctly transferred into velocity through the cross-covariances. After an initial transient of a few seconds, the gyroscope biases settle near zero and the accelerometer biases settle to small offsets, except the x-axis accelerometer bias of Dataset 9, which drifts slowly (Section 3.5).

Part 2. Velocity overlaps with the mocap, as expected. Position is not measured after initialization, but remains close to the mocap on all datasets (Figs. 4–6), since it is the integral of the corrected velocity. However, position is unobservable from velocity alone, so its error would keep growing over runs longer than these 7.5–26 s datasets.

Orientation is recovered less accurately. Roll and pitch follow the shape of the mocap but with offsets of a few hundredths of a radian (e.g. roll in Fig. 5). Yaw is the least accurate state: it deviates by up to about 0.4 rad near $t \approx 2$ s in Dataset 4 before partially recovering (Fig. 5), and has a persistent offset of about 0.07 rad in Dataset 9 (Fig. 6). This is expected, since yaw affects the predicted velocity only through the direction of the horizontal specific force, so it is observable only during horizontal acceleration.

Bias observability. The vertical accelerometer bias agrees between the two parts (about $-0.06\,\mathrm{m/s^2}$ on Dataset 1 and $0.033\,\mathrm{m/s^2}$ on Dataset 9), but the horizontal ones do not; for example, the x-axis bias on Dataset 1 is about $0.03\,\mathrm{m/s^2}$ in Part 1 but about $0.4\,\mathrm{m/s^2}$ in Part 2 (Figs. 1 and 4). A small tilt error $\delta\theta$ rotates gravity into a horizontal acceleration error of about $g\,\delta\theta$, which is indistinguishable from a horizontal accelerometer bias. In Part 1, the measured orientation separates the two; in Part 2, it cannot, so part of the tilt error is absorbed into the horizontal biases. This is consistent with the larger bias transients and the roll and pitch offsets seen in Part 2.

5 Conclusion

With pose measurements (Part 1), the filter recovers the unmeasured velocity accurately and the bias estimates converge to stable values. With velocity measurements only (Part 2), velocity and position remain accurate over the supplied runs, but orientation and the horizontal accelerometer biases are only weakly observable, with yaw the least accurate state.